Year 11 Set 4 · Foundation · Lesson 7

Straight-line graphs

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Today I will learn to...

Goals

1. Complete a table of values and plot a straight line

2. Find the gradient and y-intercept of a line

Do now: retrieval practice

4 mins

1. P(A) = 0.4. Find P(not A)

   0.6

2. Find 2x + 1 when x = 3

   7

3. Find 3x − 4 when x = −2

   −10

4. Midpoint of (2, 4) and (6, 8)

   (4, 6)

Objective 1

Complete a table of values and plot a straight line

Key idea

Learn

!

Substitute each x value into the equation. Plot the points and join them with a ruler.

Objective 1: Complete a table of values and plot a straight line

I do: example 1

I do

y = 2x + 1 for x = −1, 0, 1, 2, 3

1. Substitute each x into y = 2x + 1
2. x = −1: y = 2 × (−1) + 1 = −1
3. x = 0: y = 1 x = 1: y = 3
4. x = 2: y = 5 x = 3: y = 7

I do: example 2

I do

y = 3 − x for x = 0, 1, 2, 3

1. Substitute each x into y = 3 − x
2. x = 0: y = 3 − 0 = 3
3. x = 1: y = 2 x = 2: y = 1
4. x = 3: y = 3 − 3 = 0

I do: example 3

I do

$y = \frac{1}{2}x - 2$ for $x = -2, 0, 2, 4$

1. Substitute each x into $y = \frac{1}{2}x - 2$
2. $x = -2$: $y = \frac{1}{2} \times (-2) - 2 = -3$
3. $x = 0$: $y = -2$ $x = 2$: $y = -1$
4. $x = 4$: $y = 2 - 2 = 0$

Mini whiteboards

We do

1. $y = 3x - 1$ when $x = 2$

Answer: 5

2. $y = 3x - 1$ when $x = -1$

Answer: −4

3. $y = 2x + 5$ when $x = -3$

Answer: −1

4. $y = 4 - x$ when $x = 6$

Answer: −2

You do: choose your challenge

You do

Bronze

Find y when x = 4 for y = 2x + 3.

Silver

Complete the table for y = 5 − 2x for x = −1, 0, 1, 2.

Gold

Does the point (−2, −5) lie on the line y = 3x + 1? Explain.

Platinum

Where does y = 4x − 6 cross the x-axis?

You do: answers

Check

Bronze

11

Silver

7, 5, 3, 1

Gold

$3 \times (-2) + 1 = -5$, so yes

Platinum

$4x - 6 = 0$, $x = 1.5$, so $(1.5, 0)$

If you finish early

Extra

Bronze

Find y when x = 3 for y = 4x − 2.

10

Silver

Complete the table for y = 2x − 1 for x = −2, −1, 0, 1.

−5, −3, −1, 1

Gold

Does the point (3, 7) lie on y = 3x − 2? Explain.

3 × 3 − 2 = 7, so yes

Platinum

Where does y = 2x + 8 cross the x-axis and the y-axis?

x-axis at (−4, 0), y-axis at (0, 8)

Exam question 1

2 marks

Complete the table for $y = 4x - 3$ for $x = -1, 0, 1, 2$.

Space for working

Mark scheme: −7, −3, 1, 5

Exam question 2

1 mark

Does the point (3, 10) lie on the line y = 3x + 1? Show how you decide.

Space for working

Mark scheme: 3 × 3 + 1 = 10, so yes

Objective 1 achieved

Complete a table of values and plot a straight line



Objective 2

Find the gradient and y-intercept of a line

✓ Achieved: Complete a table of values and plot a straight line

Key idea

Learn

!

In y = mx + c, m is the gradient and c is where the line crosses the y-axis. Gradient = change in y ÷ change in x.

Objective 2: Find the gradient and y-intercept of a line

I do: example 1

I do

y = 5x + 2

1. Compare with y = mx + c
2. m is the number in front of x: m = 5
3. Gradient = 5
4. c = 2, so y-intercept = (0, 2)

I do: example 2

I do

Gradient of the line through (1, 3) and (4, 9)

1. Gradient = change in y ÷ change in x
2. Change in y: 9 − 3 = 6
3. Change in x: 4 − 1 = 3
4. Gradient = 6 ÷ 3 = 2

I do: example 3

I do

Gradient 3, crosses the y-axis at (0, −4)

1. Use $y = mx + c$
2. Gradient 3, so $m = 3$
3. Crosses the y-axis at (0, −4), so $c = -4$
4. $y = 3x - 4$

Mini whiteboards

We do

1. Gradient of $y = -2x + 7$

Answer: -2

2. Gradient through $(0, 1)$ and $(2, 9)$

Answer: 4

3. y-intercept of $y = 5x - 3$

Answer: $(0, -3)$

4. Gradient through $(1, 1)$ and $(3, -5)$

Answer: -3

You do: choose your challenge

You do

Bronze

Write down the gradient of $y = 6x - 1$.

Silver

Find the gradient of the line through (2, 3) and (5, 12).

Gold

A line has gradient -2 and passes through (0, 8). Write its equation and find where it crosses the x-axis.

Platinum

Write the equation of the line parallel to $y = 3x + 1$ that passes through (0, -2).

You do: answers

Check

Bronze

6

Silver

$9 \div 3 = 3$

Gold

$y = -2x + 8$, crosses at $(4, 0)$

Platinum

$y = 3x - 2$

If you finish early

Extra

Bronze

Write down the gradient and y-intercept of $y = 4x + 1$.

Gradient 4, intercept (0, 1)

Silver

Find the gradient of the line through (0, 3) and (4, 11).

2

Gold

A line has gradient 3 and passes through (0, −5). Write its equation.

$y = 3x - 5$

Platinum

Find the equation of the line through (1, 4) and (3, 10).

$m = 3$, $c = 1$: $y = 3x + 1$

Exam question 1

1 mark

Write down the gradient of $y = 7 - 3x$.

Space for working

Mark scheme: −3

Exam question 2

3 marks

A line passes through (0, 5) and (2, 11). Find its equation.

Space for working

Mark scheme: m = 6 ÷ 2 = 3, c = 5, y = 3x + 5

Objective 2 achieved

Find the gradient and y-intercept of a line



Real life

Apply

A plumber charges C = 25h + 40 pounds for h hours. What do 25 and 40 mean? How much for 3 hours?

£25 per hour, £40 call-out fee. 3 hours: £115

Extension: stretch yourself

Stretch

1. Find the equation of the line through (2, 5) and (4, 11).

$m = 3$, $5 = 6 + c \rightarrow c = -1$, $y = 3x - 1$

2. Are $y = 2x + 3$ and $2y = 4x - 1$ parallel?

$2y = 4x - 1 \rightarrow y = 2x - 0.5$. Both gradient 2, so yes

Exit ticket: exam questions

Exit

1. Write down the gradient and y-intercept of $y = 5x - 3$. **(2 marks)**

2. Find the gradient of the line through (1, 2) and (3, 10). **(2 marks)**

3. Find the equation of the line through (0, 1) and (2, 7). **(3 marks)**

Exit ticket: mark scheme

Mark it

Q1 Gradient 5, intercept (0, −3) [2]

Q2 8 ÷ 2 = 4 [2]

Q3 m = 3, c = 1, y = 3x + 1 [3]

Total: _____ / 7

Today I achieved

- ✔ **Complete a table of values and plot a straight line**
- ✔ **Find the gradient and y-intercept of a line**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Complete the table for $y = 2x - 5$ for $x = 0, 1, 2, 3$. [2]

2. Write down the gradient of $y = 9 - x$. [1]

3. Find the gradient of the line through $(-1, 4)$ and $(3, 12)$. [2]

4. Write the equation of the line with gradient 4 that passes through $(0, -7)$. [2]

5. Find the equation of the line through $(1, 5)$ and $(3, 11)$. [3]

Extra practice: Maths Genie → Edexcel question bank → Gradient of a Line · Equation of a Line